

C:\Users\Admin\OneDrive - South East Technological University\12. Paper 4\Post workshop\Element 2 \Result\Element2_PathAnalysis_Thi Minh Thuan Phung_20115516.amw

Analysis Summary

Date and Time

Date: Wednesday, July 29, 2026

Time: 6:43:56 PM

Title

Element2_pathanalysis_thi minh thuan phung_20115516: Wednesday, July 29, 2026 6:43 PM

Groups

Group number 1 (Group number 1)

Notes for Group (Group number 1)

The model is recursive.

Sample size = 100

Variable Summary (Group number 1)

Your model contains the following variables (Group number 1)

Observed, endogenous variables

ID5
ID4
ID3
ID2
ID1
OC2
OC1
SI4
SI3
SI2

SI1
 ES4
 ES3
 ES2
 ES1
 HB4
 HB3
 HB2
 HB1

Unobserved, endogenous variables

Investment_Decision

Unobserved, exogenous variables

e1
 e2
 e3
 e4
 e5
 Overconfidence
 e6
 e7
 Social_influence
 e8
 e9
 e10
 e11
 Emotional_State
 e12
 e13
 e14
 e15
 Herding_Behaviour
 e16
 e17
 e18
 e19
 e20

Variable counts (Group number 1)

Number of variables in your model: 44
 Number of observed variables: 19
 Number of unobserved variables: 25
 Number of exogenous variables: 24
 Number of endogenous variables: 20

Parameter Summary (Group number 1)

	Weights	Covariances	Variances	Means	Intercepts	Total
Fixed	25	0	0	0	0	25
Labeled	0	0	0	0	0	0
Unlabeled	18	8	24	0	0	50
Total	43	8	24	0	0	75

Models

Default model (Default model)

Notes for Model (Default model)

Computation of degrees of freedom (Default model)

Number of distinct sample moments: 190

Number of distinct parameters to be estimated: 50

Degrees of freedom (190 - 50): 140

Result (Default model)

Minimum was achieved

Chi-square = 301.388

Degrees of freedom = 140

Probability level = .000

Group number 1 (Group number 1 - Default model)

Estimates (Group number 1 - Default model)

Scalar Estimates (Group number 1 - Default model)

Maximum Likelihood Estimates

Regression Weights: (Group number 1 - Default model)

		Estimate	S.E.	C.R.	PLabel
Investment_Decision	<--- Overconfidence	.496	.170	2.914	.004
Investment_Decision	<--- Herding_Behaviour	.068	.119	.574	.566
Investment_Decision	<--- Emotional_State	.468	.174	2.699	.007
Investment_Decision	<--- Social_influence	-.085	.188	-.452	.652
ID5	<--- Investment_Decision	1.000			
ID4	<--- Investment_Decision	.989	.130	7.628	***
ID2	<--- Investment_Decision	.707	.111	6.375	***

ID1	<---	Investment_Decision	.623	.116	5.377	***
OC2	<---	Overconfidence	.729	.169	4.315	***
OC1	<---	Overconfidence	1.000			
SI4	<---	Social_influence	1.000			
SI3	<---	Social_influence	1.099	.139	7.903	***
SI2	<---	Social_influence	.853	.128	6.692	***
SI1	<---	Social_influence	.702	.116	6.073	***
ES4	<---	Emotional_State	1.000			
ES3	<---	Emotional_State	1.053	.134	7.868	***
ES2	<---	Emotional_State	.671	.108	6.202	***
ES1	<---	Emotional_State	.675	.121	5.591	***
HB4	<---	Herding_Behaviour	1.000			
HB3	<---	Herding_Behaviour	.961	.110	8.709	***
HB2	<---	Herding_Behaviour	1.050	.112	9.380	***
HB1	<---	Herding_Behaviour	1.127	.115	9.776	***
ID3	<---	Investment_Decision	.613	.112	5.472	***

Standardized Regression Weights: (Group number 1 - Default model)

			Estimate
Investment_Decision	<---	Overconfidence	.436
Investment_Decision	<---	Herding_Behaviour	.065
Investment_Decision	<---	Emotional_State	.466
Investment_Decision	<---	Social_influence	-.079
ID5	<---	Investment_Decision	.852
ID4	<---	Investment_Decision	.748
ID2	<---	Investment_Decision	.637
ID1	<---	Investment_Decision	.548
OC2	<---	Overconfidence	.665
OC1	<---	Overconfidence	.887
SI4	<---	Social_influence	.787
SI3	<---	Social_influence	.829
SI2	<---	Social_influence	.696
SI1	<---	Social_influence	.638
ES4	<---	Emotional_State	.780
ES3	<---	Emotional_State	.841
ES2	<---	Emotional_State	.650
ES1	<---	Emotional_State	.591
HB4	<---	Herding_Behaviour	.813
HB3	<---	Herding_Behaviour	.792
HB2	<---	Herding_Behaviour	.839
HB1	<---	Herding_Behaviour	.869
ID3	<---	Investment_Decision	.557

Covariances: (Group number 1 - Default model)

		Estimate	S.E.	C.R.	P Label
Overconfidence	<--> Social_influence	.130	.074	1.750	.080
Overconfidence	<--> Herding_Behaviour	.197	.075	2.609	.009
Overconfidence	<--> Emotional_State	.057	.078	.729	.466
Social_influence	<--> Herding_Behaviour	-.073	.075	-.977	.328
Social_influence	<--> Emotional_State	.468	.107	4.354	***
Emotional_State	<--> Herding_Behaviour	-.055	.080	-.682	.495
e14	<--> e15	.131	.063	2.086	.037
e10	<--> e11	.123	.060	2.041	.041

Correlations: (Group number 1 - Default model)

	Estimate
Overconfidence <--> Social_influence	.219
Overconfidence <--> Herding_Behaviour	.325
Overconfidence <--> Emotional_State	.089
Social_influence <--> Herding_Behaviour	-.114
Social_influence <--> Emotional_State	.693
Emotional_State <--> Herding_Behaviour	-.080
e14 <--> e15	.252
e10 <--> e11	.261

Variances: (Group number 1 - Default model)

	Estimate	S.E.	C.R.	P Label
Overconfidence	.562	.152	3.704	***
Social_influence	.632	.145	4.349	***
Emotional_State	.721	.168	4.288	***
Herding_Behaviour	.654	.138	4.745	***
e20	.434	.107	4.049	***
e1	.275	.072	3.800	***
e2	.559	.103	5.414	***
e3	.607	.093	6.493	***
e4	.531	.086	6.200	***
e5	.656	.101	6.518	***
e6	.375	.081	4.634	***
e7	.153	.117	1.308	.191
e8	.388	.079	4.945	***
e9	.348	.082	4.241	***
e10	.490	.085	5.783	***
e11	.453	.074	6.083	***
e12	.465	.093	5.020	***

e13	.331	.083	3.971	***
e14	.444	.073	6.079	***
e15	.611	.097	6.313	***
e16	.336	.061	5.482	***
e17	.358	.063	5.698	***
e18	.303	.059	5.121	***
e19	.270	.059	4.564	***

Squared Multiple Correlations: (Group number 1 - Default model)

	Estimate
Investment_Decision	.403
HB1	.755
HB2	.704
HB3	.628
HB4	.661
ES1	.350
ES2	.422
ES3	.707
ES4	.608
SI1	.407
SI2	.484
SI3	.687
SI4	.619
OC1	.786
OC2	.443
ID1	.301
ID2	.406
ID3	.310
ID4	.560
ID5	.725

Modification Indices (Group number 1 - Default model)

Covariances: (Group number 1 - Default model)

	M.I. Par Change
e15 <--> Overconfidence	15.851 .239
e14 <--> Overconfidence	7.345 -.141
e14 <--> e20	5.316 .121
e14 <--> e19	5.051 -.094
e14 <--> e17	4.123 .089
e13 <--> e18	4.703 -.098

e11 <--> e16	4.118	-.088
e10 <--> Overconfidence	5.253	.126
e10 <--> e20	7.390	-.152
e8 <--> e20	4.721	.122
e8 <--> e14	6.753	.124
e7 <--> e19	4.041	-.080
e7 <--> e15	10.013	.157
e7 <--> e13	7.910	-.129
e7 <--> e8	8.995	-.137
e6 <--> e20	4.686	-.112
e6 <--> e19	6.060	.101
e6 <--> e17	4.048	.087
e6 <--> e16	4.072	-.086
e6 <--> e13	7.467	.129
e6 <--> e10	4.777	.102
e5 <--> Emotional_State	4.610	-.136
e5 <--> Social_influence	5.462	.135
e5 <--> e9	6.486	.153
e4 <--> Herding_Behaviour	4.330	.130
e4 <--> e9	6.222	.137
e3 <--> e20	5.825	-.152
e3 <--> e12	8.858	.187
e3 <--> e4	4.463	.131
e2 <--> e13	6.117	.147
e1 <--> e12	4.384	-.108
e1 <--> e3	9.283	-.161

Variances: (Group number 1 - Default model)

	M.I. Par Change
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Regression Weights: (Group number 1 - Default model)

	M.I. Par Change
HB2 <--- ID5	4.543 .135
HB4 <--- Emotional_State	5.369 -.191
HB4 <--- Social_influence	6.399 -.223
HB4 <--- ES3	5.625 -.144
HB4 <--- ES4	4.570 -.127
HB4 <--- SI1	9.264 -.226
HB4 <--- SI4	6.343 -.161
HB4 <--- OC2	5.603 -.187
ES1 <--- Overconfidence	14.036 .426
ES1 <--- OC1	15.375 .360
ES1 <--- OC2	7.595 .261

ES2 <---	Overconfidence	4.943	-.219
ES2 <---	OC1	5.299	-.183
ES2 <---	OC2	7.339	-.222
ES3 <---	HB2	5.750	-.170
ES3 <---	HB4	4.024	-.145
ES3 <---	OC1	4.161	-.173
ES4 <---	HB2	4.052	.155
ES4 <---	ID3	4.584	.178
SI2 <---	Overconfidence	6.254	.261
SI2 <---	OC1	5.909	.205
SI2 <---	OC2	9.200	.263
SI3 <---	ID1	5.131	.168
SI3 <---	ID2	4.247	.156
SI4 <---	ES2	5.553	.193
SI4 <---	OC1	5.260	-.194
OC1 <---	ES3	6.435	-.153
OC1 <---	SI4	8.202	-.182
OC2 <---	Emotional_State	5.662	.201
OC2 <---	Social_influence	5.861	.218
OC2 <---	ES3	9.261	.189
OC2 <---	SI2	9.393	.208
OC2 <---	SI4	5.516	.154
ID1 <---	OC2	4.418	-.215
ID3 <---	Herding_Behaviour	4.373	.220
ID3 <---	Overconfidence	8.154	.339
ID3 <---	HB3	4.927	.183
ID3 <---	ES4	5.570	.175
ID3 <---	OC1	8.181	.274
ID3 <---	OC2	5.493	.231
ID4 <---	Herding_Behaviour	5.834	-.261
ID4 <---	HB1	5.004	-.177
ID4 <---	HB2	4.259	-.169
ID4 <---	HB3	4.910	-.188
ID4 <---	ES3	4.723	.170
ID4 <---	OC1	4.306	-.204
ID5 <---	ES4	5.441	-.143
ID5 <---	ID3	6.242	-.178

Minimization History (Default model)

Iteration	Negative eigenvalues	Condition #	Smallest eigenvalue	Diameter	FN	Tries	Ratio
0	e	13	-.558	9999.000	1117.994	0	9999.000
1	e*	6	-.084	3.164	520.371	20	.493
2	e	2	-.021	.754	386.427	6	.887

3	e	0	131.357	.704	322.637	5	.857
4	e	0	213.653	1.025	314.048	2	.000
5	e	0	351.156	.300	303.220	1	1.132
6	e	0	265.307	.160	301.575	1	1.171
7	e	0	271.541	.056	301.392	1	1.085
8	e	0	280.669	.009	301.388	1	1.014
9	e	0	285.067	.000	301.388	1	1.000

Model Fit Summary

CMIN

Model	NPAR	CMIN	DF	P	CMIN/DF
Default model	50	301.388	140	.000	2.153
Saturated model	190	.000	0		
Independence model	19	1110.971	171	.000	6.497

RMR, GFI

Model	RMR	GFI	AGFI	PGFI
Default model	.084	.777	.697	.572
Saturated model	.000	1.000		
Independence model	.285	.372	.303	.335

Baseline Comparisons

Model	NFI Delta1	RFI rho1	IFI Delta2	TLI rho2	CFI
Default model	.729	.669	.834	.790	.828
Saturated model	1.000		1.000		1.000
Independence model	.000	.000	.000	.000	.000

Parsimony-Adjusted Measures

Model	PRATIO	PNFI	PCFI
Default model	.819	.597	.678
Saturated model	.000	.000	.000
Independence model	1.000	.000	.000

NCP

Model	NCP	LO 90	HI 90
Default model	161.388	115.175	215.350

Saturated model	.000	.000	.000
Independence model	939.971	838.324	1049.092

FMIN

Model	FMIN	F0	LO 90	HI 90
Default model	3.044	1.630	1.163	2.175
Saturated model	.000	.000	.000	.000
Independence model	11.222	9.495	8.468	10.597

RMSEA

Model	RMSEA	LO 90	HI 90	PCLOSE
Default model	.108	.091	.125	.000
Independence model	.236	.223	.249	.000

AIC

Model	AIC	BCC	BIC	CAIC
Default model	401.388	426.704	531.647	581.647
Saturated model	380.000	476.203	874.982	1064.982
Independence model	1148.971	1158.591	1198.469	1217.469

ECVI

Model	ECVI	LO 90	HI 90	MECVI
Default model	4.054	3.588	4.599	4.310
Saturated model	3.838	3.838	3.838	4.810
Independence model	11.606	10.579	12.708	11.703

HOELTER

Model	HOELTER	HOELTER
	.05	.01
Default model	56	60
Independence model	19	20

Execution time summary

Minimization: .034
Miscellaneous: .303
Bootstrap: .000
Total: .337